3 A sound system in a large auditorium has two identical loudspeakers, S_1 and S_2, separated by 1.5 m. The amplitude of the sound emitted by each loudspeaker is proportional to the voltage across its terminals. A balance control is used to adjust the voltage supplied to each loudspeaker. The arrangement is shown in Fig. 3.1.

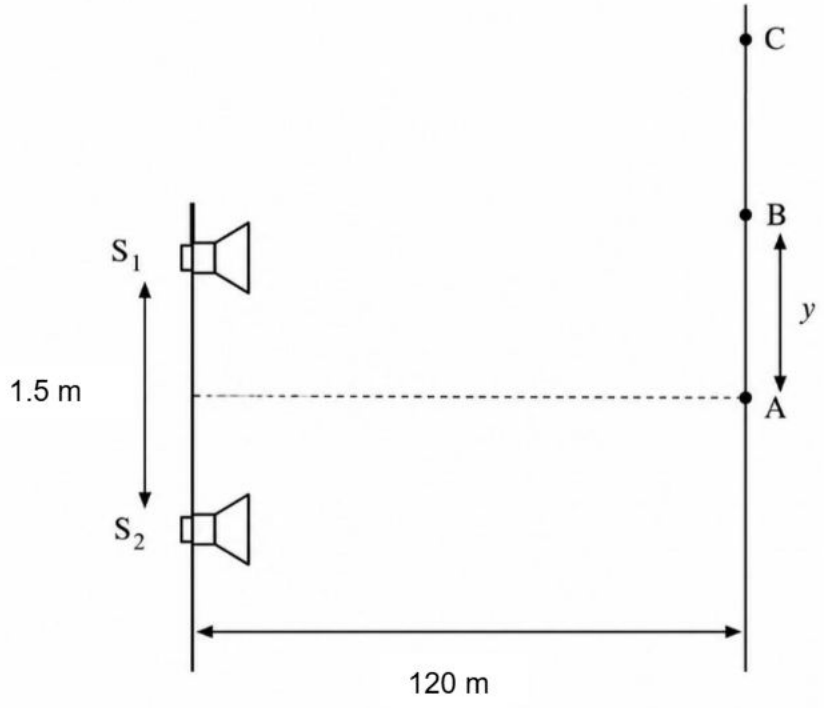


Fig. 3.1 (not to scale)

Initially, the loudspeakers emit sound waves of frequency 8500 Hz in phase. The balance control is adjusted so that the potential difference across each loudspeaker is 5.0 V r.m.s.

An observer at A hears a loud sound of intensity I_{max}. As he moves along the line AC, which is 120 m from the loudspeakers, the intensity first falls to zero at B, a distance y from A.

The speed of sound in air is 340 m s^{-1}.

(a) Determine the distance y.

$$\lambda = \frac{v}{f} = \frac{340}{8500} = 0.0400\ \text{m} \qquad [1]$$

Let the distance between the first two maxima be x.

$$x = \frac{\lambda D}{D} = \frac{0.0400 \times 120}{1.5} = 3.2\ \text{m} \quad [1]$$

Since B is the first minimum, the distance y is half of x.

$$y = \frac{x}{2} = \frac{3.2}{2} = 1.6\ \text{m} \quad [1]$$

y = ... m [3]

(b) Determine the next higher frequency of sound emitted by the speaker such that point B would also be a point of zero intensity.

The path difference at B is $\Delta x = \frac{\lambda}{2} = \frac{0.0400}{2} = 0.0200\,\text{m}$ [1]

For destructive interference, $\Delta x = (n + \frac{\lambda}{2})$

For the original frequency, this corresponds to n = 0. If the frequency is increased, it will correspond to n = 1 or $\Delta x = \frac{3\lambda'}{2}$

Hence

$0.0200 = \frac{3\lambda'}{2}$

$\lambda' = 0.01333\,\text{m}$ [1]

$f' = \frac{v}{\lambda'} = \frac{340}{0.01333} = 2.55 \times 10^4\,\text{Hz}$ [1]

frequency = ... Hz [3]

(c) The loudspeakers continue to emit sound of frequency 8500 Hz. The balance control is adjusted so that the voltages applied across S_1 and S_2 are 2.0 V r.m.s. and 8.0 V r.m.s. respectively. The amplitude of the sound emitted by each loudspeaker is proportional to the voltage applied across it.

In terms of I_{max}, determine the new intensity at B.

Initially, resultant amplitude at A, $A = A_1 + A_2 = 5 + 5 = 10$ units [1]

Later, the balance control is changed. At B, the waves meet out of phase:
$\therefore A' = A_1 - A_2 = 8 - 2 = 6$ units [1]

$\frac{I_{max}'}{I_{max}} = \frac{kA'^2}{kA^2} = (\frac{6}{10})^2$

$\therefore I_{max}' = 0.36 I_{max}$ [1]

Intensity = ...I_{max} [3]

Total: [9]